HALIÇ UNIVERSITY / FACULTY OF ENGINEERING / DEPARTMENT OF COMPUTER ENGINEERING
2024 - 2025 ACADEMIC YEAR / SPRING SEMESTER / CENG01 – NUMERICAL ANALYSIS
MIDTERM EXAM - 25/03/2025

Name & Surname :

Student Number :

Signature :

INSTRUCTIONS:

- The exam will start at 13:00 and will finish at 14:50. The total time of the exam is **90** minutes.
- This exam consists of six questions. Please explain your answers step by step.
- Write **four** digits after the comma/point.
- Use **radian** when calculating with **trigonometric** functions.

Good luck!

QUESTIONS

Q1. Let $\tilde{x}$ be some approximation of x. Please define the absolute error and relative error. Why might we need to use the relative error? Explain with a simple example that we use relative error in numerical calculations to show that the approximation is more meaningful/better. (10 pts)

Answer:

Q2. Let $f(x) = e^x \sin(x)$ and $x_0 = 0$.

a. Determine the Taylor series expansion about x_0 with the first three terms. (10 pts)

b. Calculate the approximate value of the $f(x)$ by using $x = 0.1$ in the expansion you obtained. (5 pts)

Answer:

Q3. Consider the equation $4.5x - 2\cos(x) = 0$.

a. Prove that this has at least one real root in the interval $[0, 1]$. (5 pts)
b. Calculate the approximate root of the given equation within the interval $[0, 1]$ using the *Regula-Falsi* method after three iterations. (15 pts)

Answer:

Q4. The irrational number $\sqrt{2}$ can be approximated by applying the Newton's method to solve the nonlinear equation $x^2 - 2 = 0$. Write the iteration formula. (15 pts)
Answer:



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Q6. Consider the following system of four linear equations:

$$\begin{aligned} 10x_1 - x_2 + 2x_3 &= 6 \\ -x_1 + 11x_2 - x_3 + 3x_4 &= 25 \\ 2x_1 - x_2 + 10x_3 - x_4 &= -11 \\ 3x_2 - x_3 + 8x_4 &= 15 \end{aligned}$$

a. Create the iterative system in detail using Jacobi iterative method. (10 pts)

b. Carry out the first two iterations of the solution of this system using this method. (10 pts)

Answer: